Memphis Zoo Custodial System

Operational design, screen walkthrough, and cost comparison

A practical view of how the custodial system connects daily area ownership, scan-based work confirmation, guest issue reporting, events, reminders, messaging, and Memphis AI into one working operations flow.

01 See the work

Managers can see scan activity, completion status, guest issues, and schedules from the screens they already use.

02 Guide the day

Custodians get simple phone workflows: assigned areas, scan prompts, event notices, due soon reminders, and issue notifications.

03 Keep proof simple

NFC/QR activity creates a time-stamped trail without asking employees to manage spreadsheets or paperwork.

04 Control cost

The design uses low-cost phones, NFC tags, kiosk licenses, GitHub Pages, Supabase, Render, and zoo Wi-Fi instead of broad commercial software licensing.

Important framing

The system does not replace manager inspections. It tells managers when and where cleaned areas are ready to inspect, so checks can happen before guest traffic re-soils the area.

Document style

This version uses the normal print/PC page layout and full application screenshots. It is not a phone-friendly image deck.

How the System Works

The core idea is simple: a scan or report becomes a visible operations signal, and the right person gets notified.

1. Area ownership

Schedules assign custodians to restroom, exhibit, and service areas for the day.

2. Work starts at the location

The custodian uses the phone workflow and NFC/QR scan points to identify the real location.

3. Local-first activity

If Wi-Fi is weak, the phone keeps working locally and queues the action.

4. Sync and visibility

When connection returns, queued activity syncs to the backend and dashboard views refresh.

5. Messenger notifications

Guest issues, event notices, due soon locations, and overdue locations create Memphis Messenger alerts.

6. Manager inspection

Managers use the screens to know what is ready to inspect; the software supports the inspection, it does not replace it.

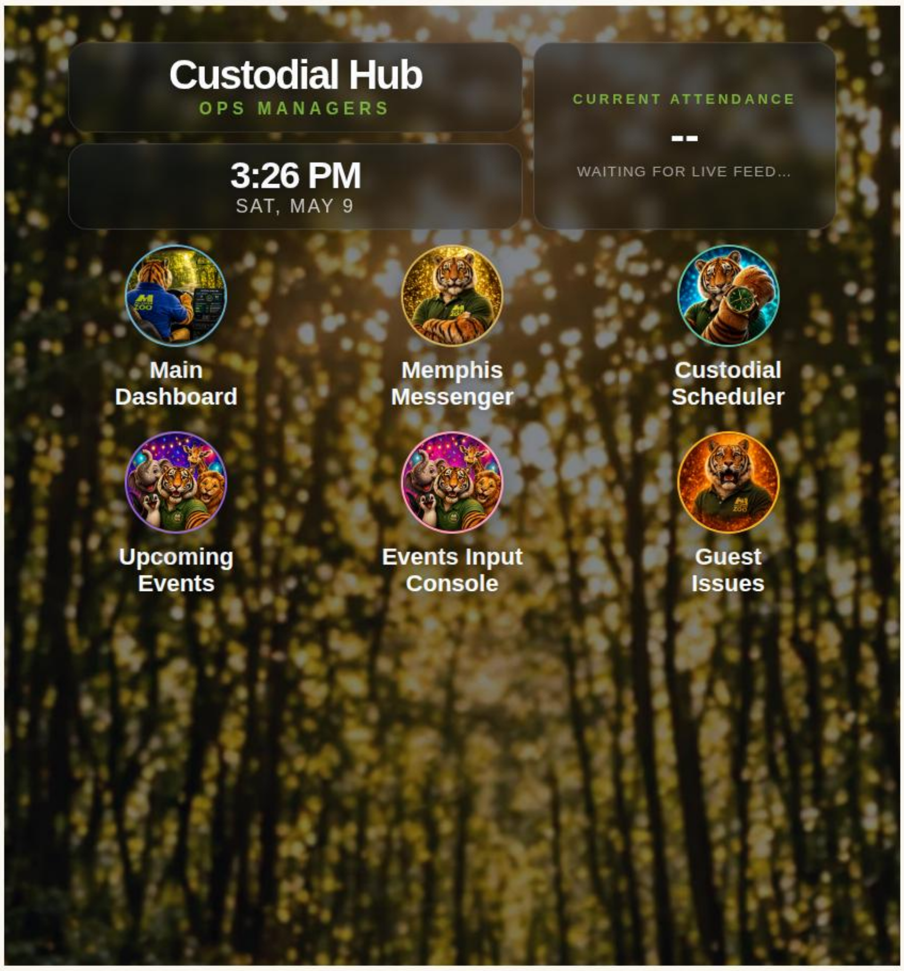
The system is strongest because the pieces are connected. Commercial tools often handle inspections, work orders, QR requests, or schedules separately. This design ties the custodial workflow together around the actual zoo day: staffing, locations, guests, events, scans, messages, and manager follow-up.

WHAT CHANGES OPERATIONALLY

- Less guessing about which areas are ready for inspection.
- Less dependence on radio calls and memory when mornings get busy.
- Clearer response path when a guest reports a cleanliness issue.
- Event information reaches the people assigned to the affected area.
- Due soon and overdue reminders help prevent quiet misses.

Ops Manager Hub

PC landing view for the operations side of the custodial system.

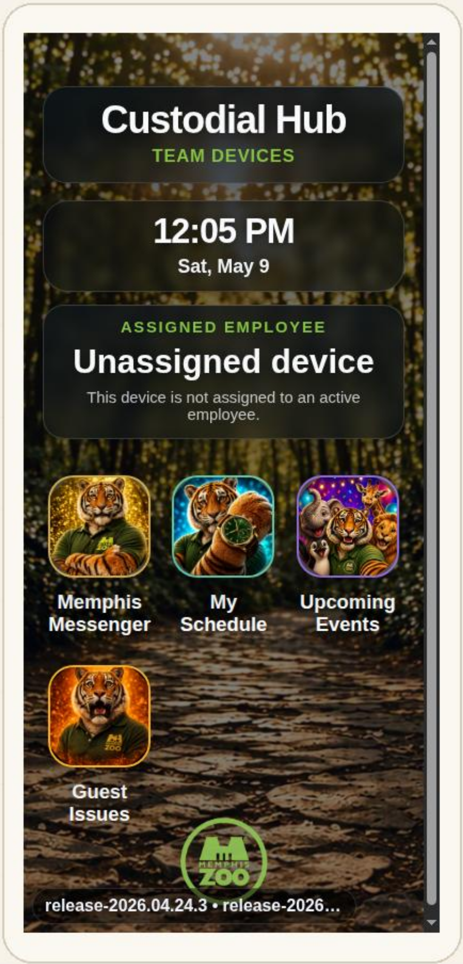


WHAT THIS SCREEN SHOWS

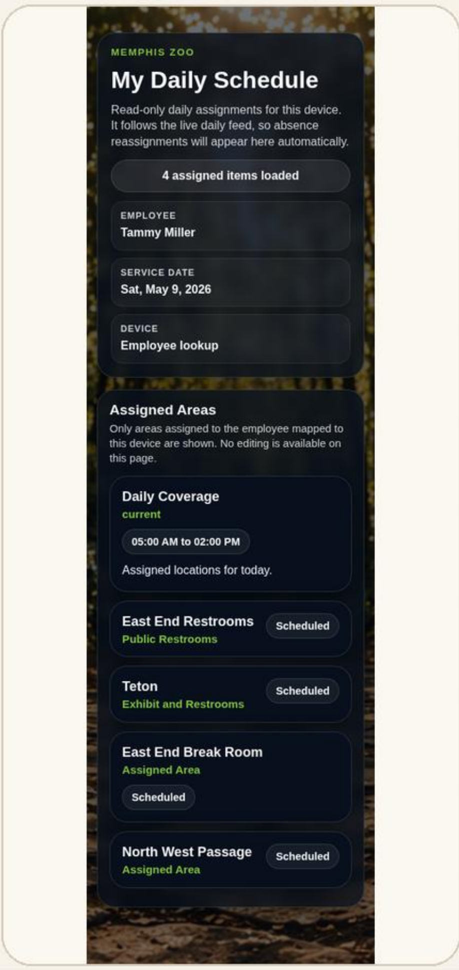
- The hub gives managers one clean starting point for the major operational screens.
- Dashboard, schedules, guest issues, events, messages, and system tools are grouped from one landing view.
- The attendance panel gives quick staff visibility without opening a separate report first.
- This keeps the manager workflow simple: start at the hub, then open the screen needed for the day.

Custodian Phone Views

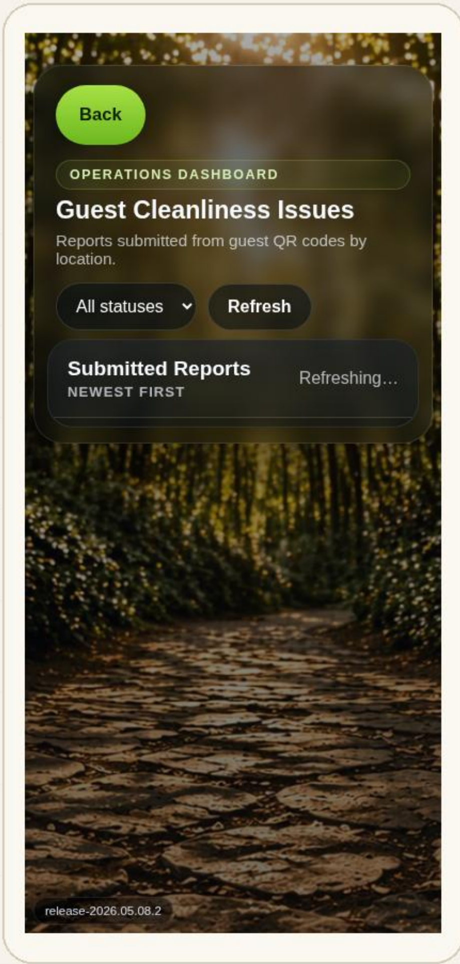
The phone side is intentionally simple: open the hub, choose the task, scan or report, and keep moving.



Employee Hub



Tammy Miller — My Schedule



Guest Issues View

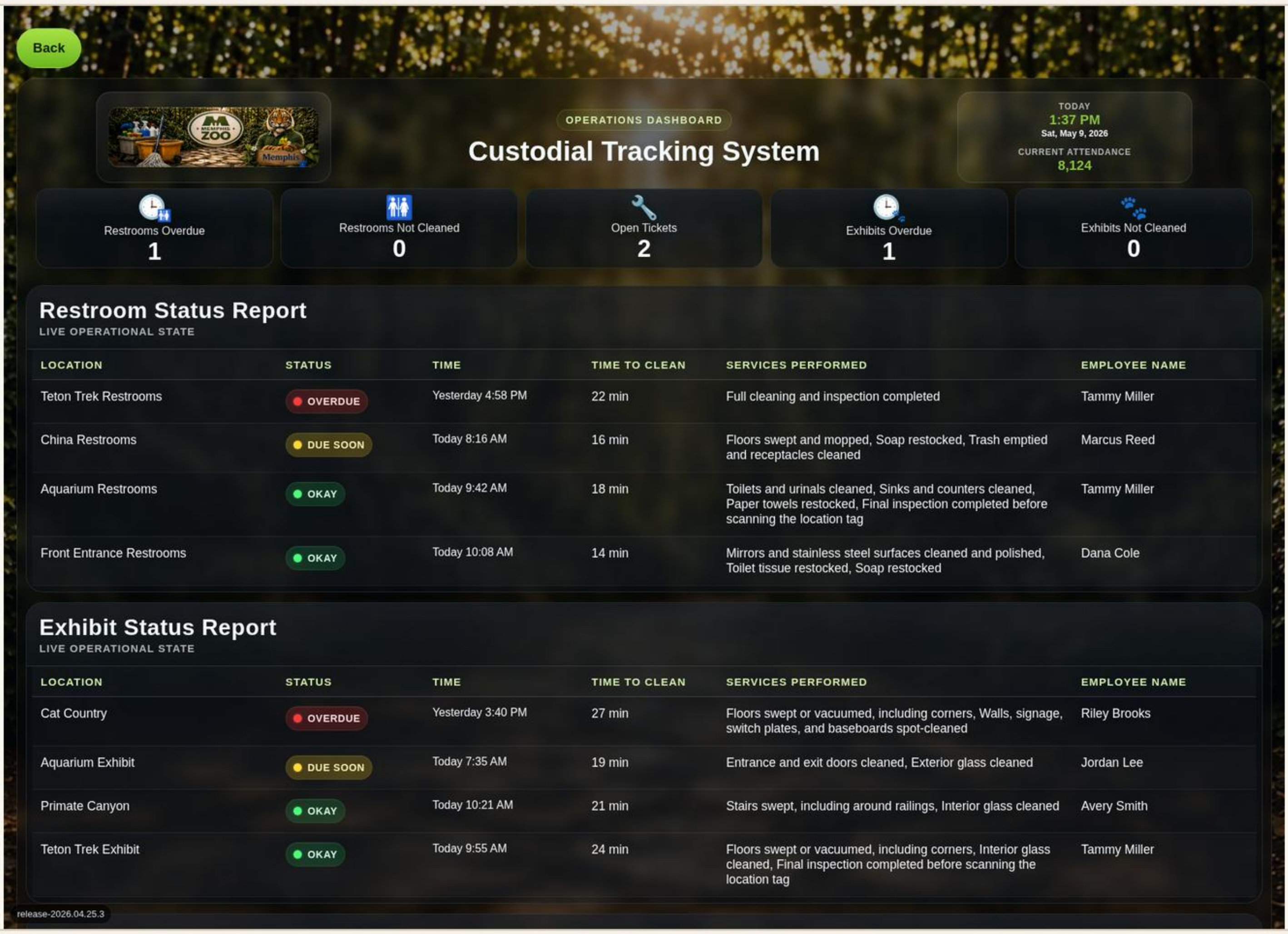
HOW THE PHONE WORKFLOW SHOULD FEEL

- Custodians do not need to understand the backend. They see their schedule, assigned areas, messages, events, and guest issues.
- Tammy’s schedule shows assigned locations such as Teton as Exhibit and Restrooms when both workflows apply.
- A guest QR issue becomes a Memphis Messenger notification so the assigned custodian and operations contact can respond.
- The same notification pattern supports events in the employee’s area, due soon reminders, and overdue location reminders.
- The workflow is built for zoo Wi-Fi. If connection drops, work can continue locally and sync when service returns.

> **Design target**
> The phone screens are for quick action in the field. The PC screens are for planning, visibility, and follow-up.

Operations Dashboard

PC view for daily visibility: cleaned areas, scan activity, open issues, and inspection readiness.

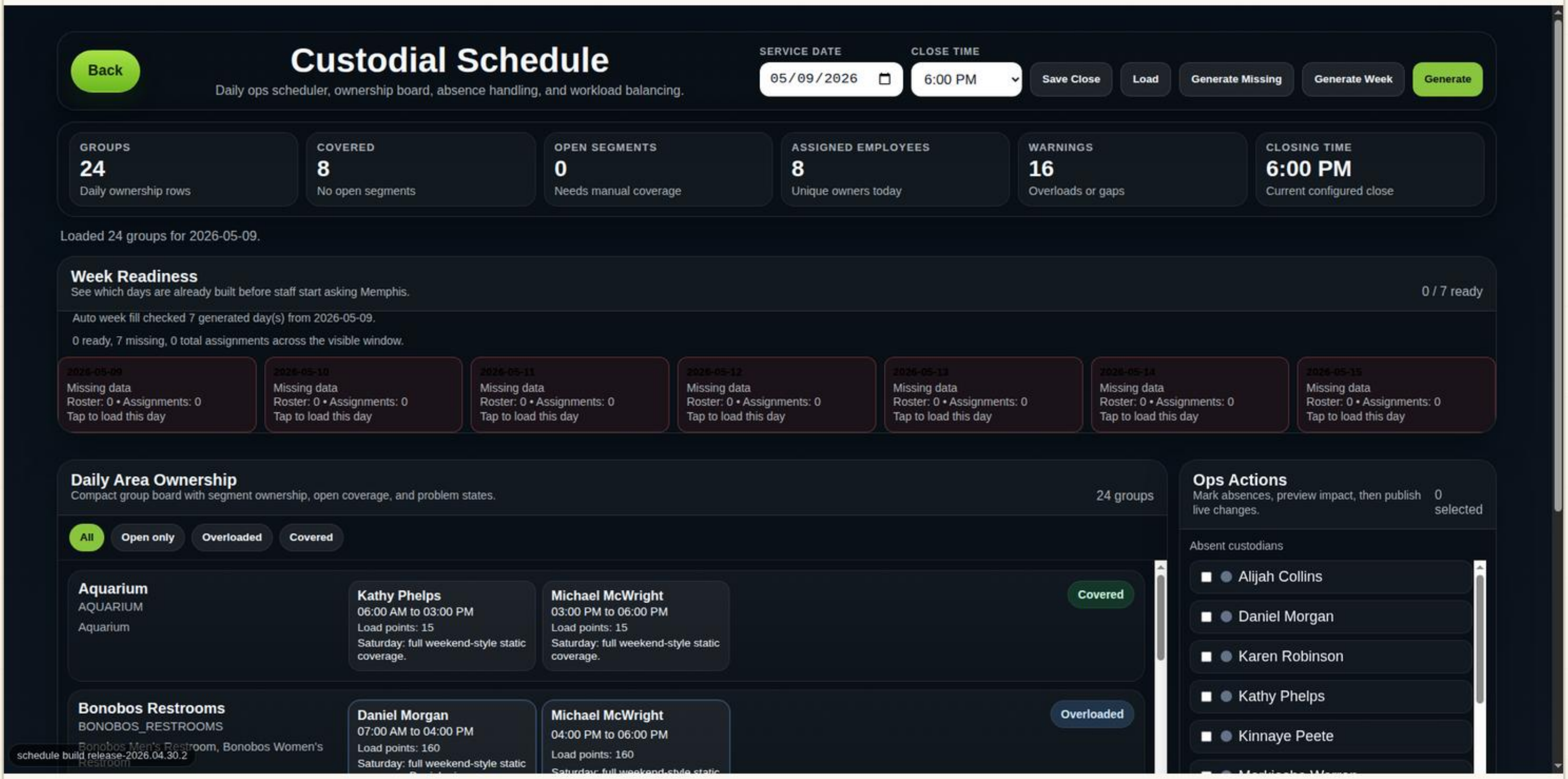


WHAT THIS SCREEN SHOWS

- Shows the manager-facing control view rather than asking someone to track restroom/exhibit status manually.
- Refresh behavior keeps the screen current while the day is moving. The screenshot is used as a screen example, not a claim that rollout is complete everywhere.
- The goal is not to replace inspection; it helps managers decide where inspection attention should go next.
- The value is quick visibility: what is active, what needs follow-up, and what is ready for a manager to check.

Daily Schedule and Absence Reassignment

PC view for area ownership, close time, absence handling, and coverage planning.

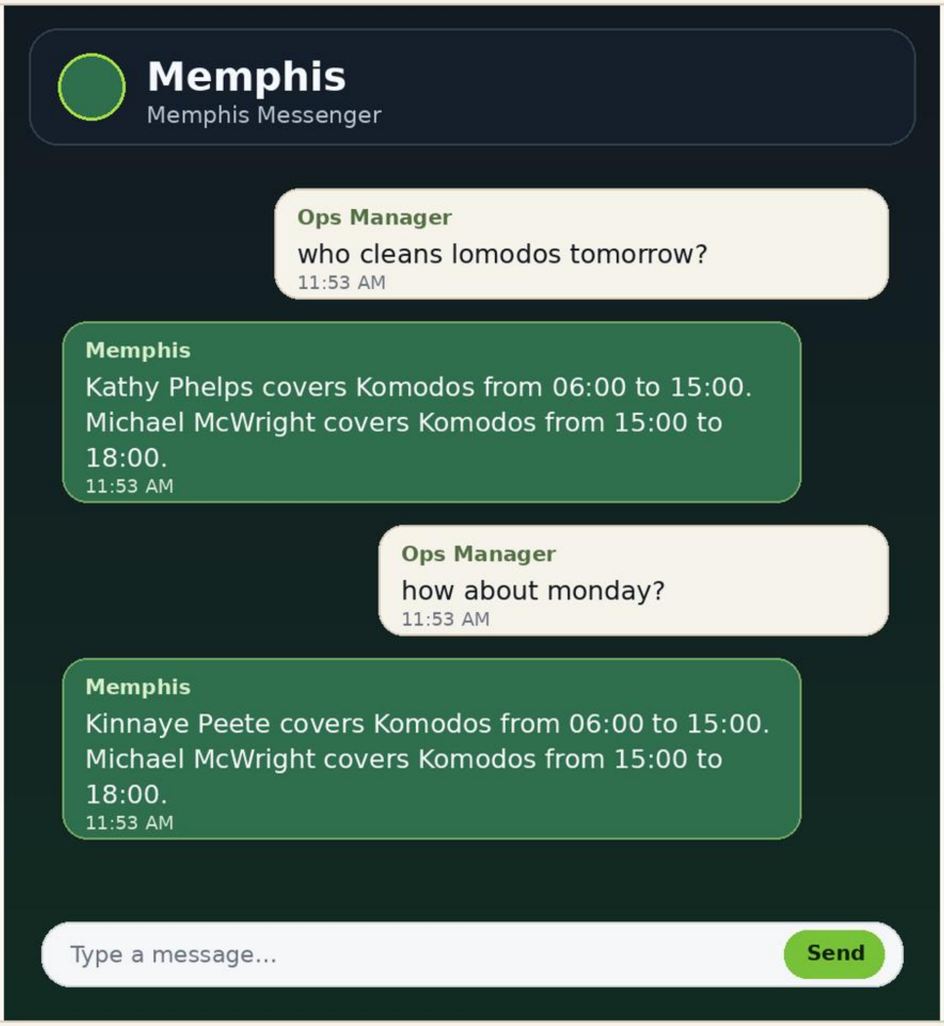


WHAT THIS SCREEN SHOWS

- Daily area ownership connects employees to location groups before the day begins.
- Absence tools preview and apply reassignment instead of leaving coverage decisions buried in notes or memory.
- Close-time and workload settings give the system context for reminders and coverage expectations.
- This keeps coverage decisions tied to the same system that handles scan activity and reminders.

Memphis Messenger

PC view showing operational messages inside the Messenger conversation.

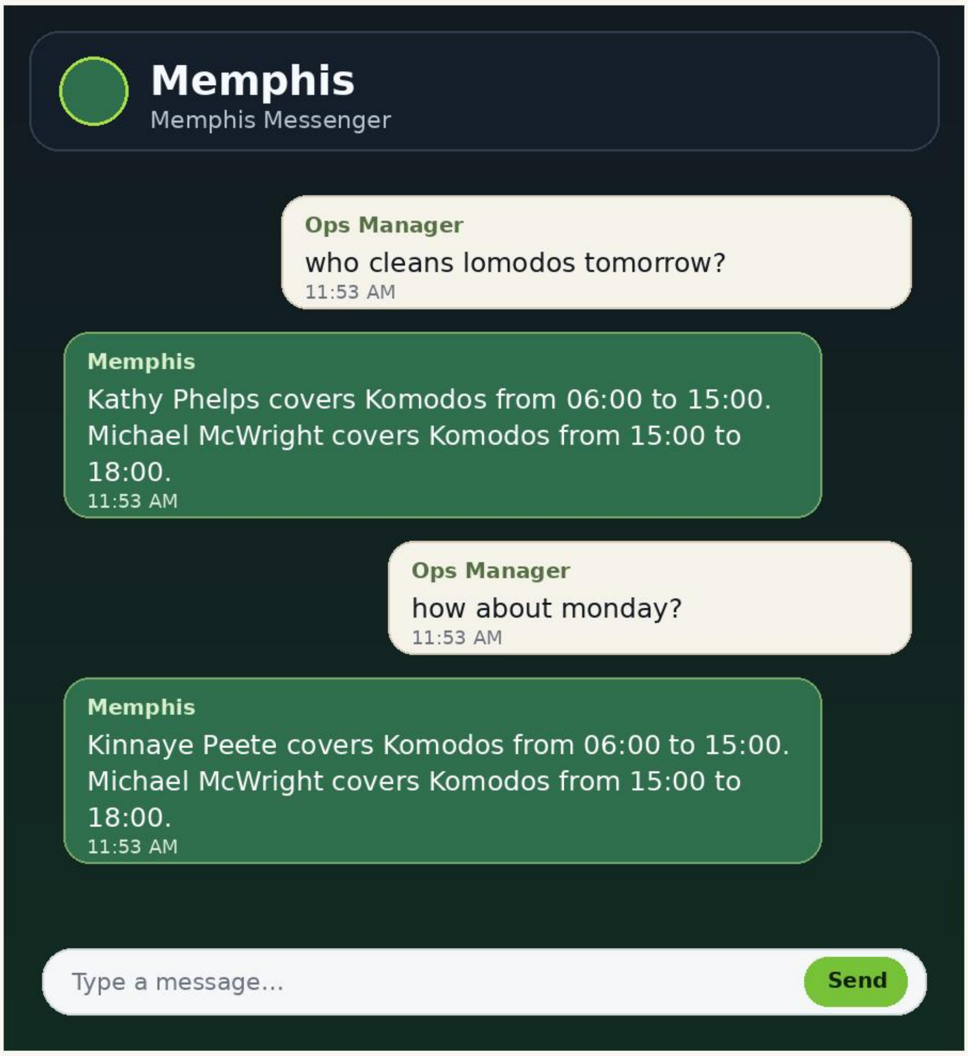


WHAT THIS SCREEN SHOWS

- Messenger is the notification path for operational prompts, not just general chat.
- Guest issues can trigger Memphis Messenger notifications to the assigned custodian and operations contact.
- Events in an assigned area, due soon locations, and overdue locations follow the same notification concept.
- The visible examples show Memphis keeping the conversation context as the manager asks follow-up questions.

Memphis AI inside Memphis Messenger

The Memphis AI examples are shown as live message bubbles inside the Messenger conversation, not as a separate mockup.

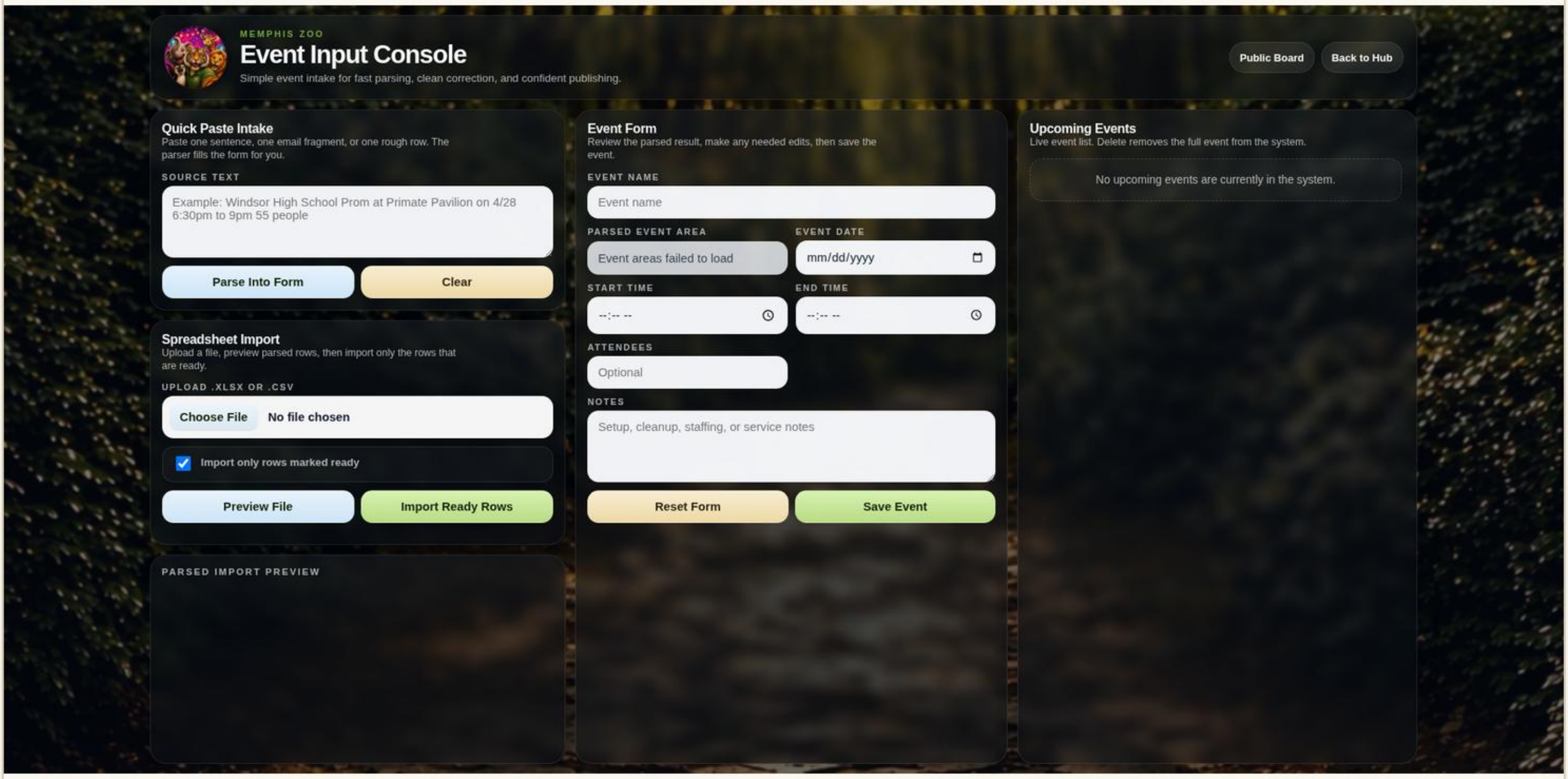


WHAT MEMPHIS ADDS

- Answers operational questions from system data: staffing, schedules, area coverage, tickets, scans, events, attendance, and contacts.
- Keeps routine answers inside Memphis Messenger so a manager can ask follow-up questions in the same place notifications arrive.
- Supports notification workflows: guest issue reported, event in assigned area, due soon location, and overdue location.
- Memphis can handle simple misspellings: the example shows “lomodos” being understood as the intended Komodos location.

Event Input Console

PC view for entering events by paste, spreadsheet import, or manual form.

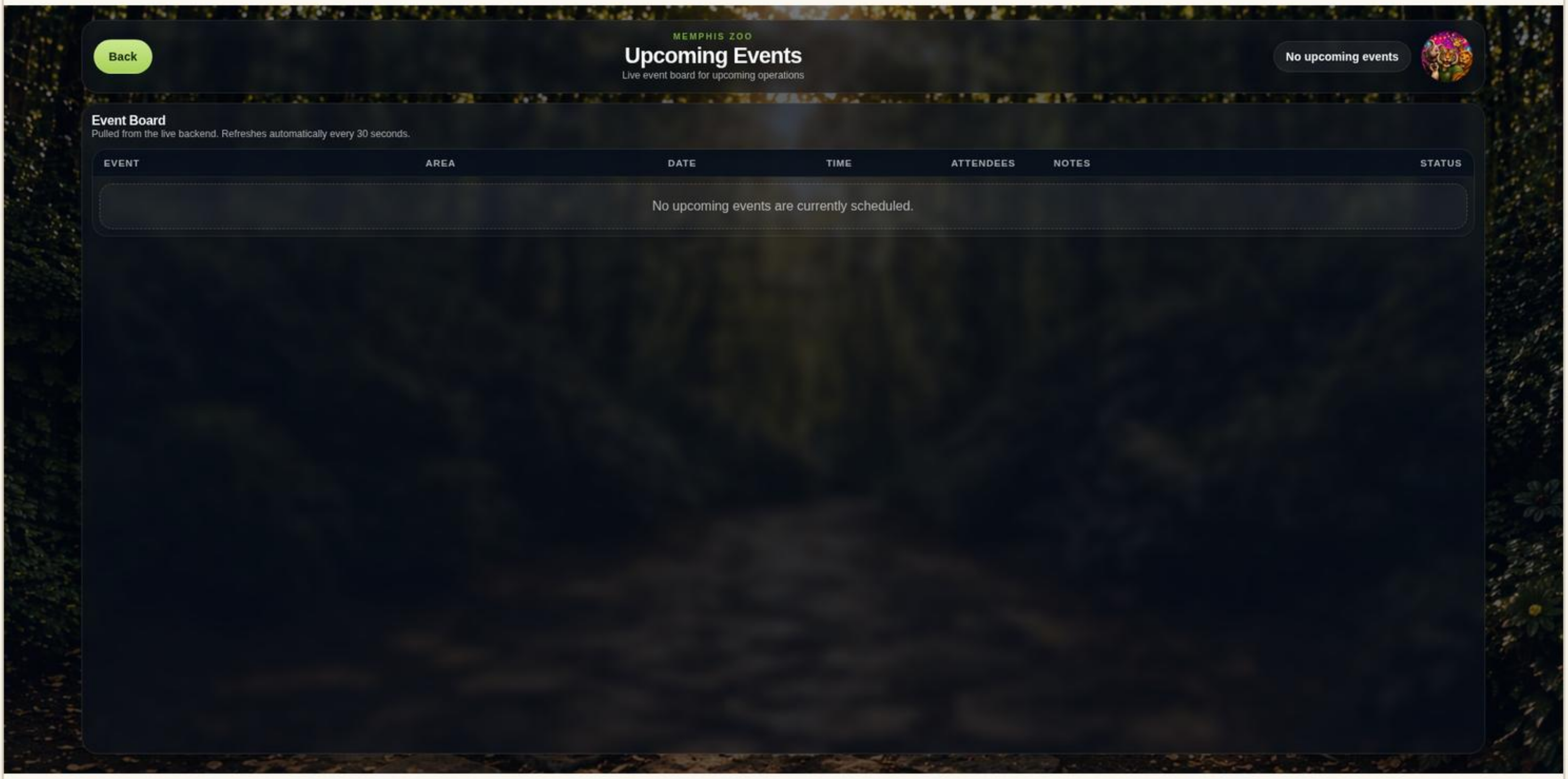


WHAT THIS SCREEN SHOWS

- Staff can paste copied event rows, import spreadsheet-style information, or use the form. The console pulls out the event name, date, time, location, area, and notes needed for custodial planning.
- The event area matters because assigned custodians can be notified when something affects their zone.
- Extra columns or unrelated details can be ignored, so event information becomes useful without manually cleaning every spreadsheet first.
- That enables practical Memphis Messenger prompts such as “events in your area today.”

Upcoming Events Display

PC view for the clean event display format.



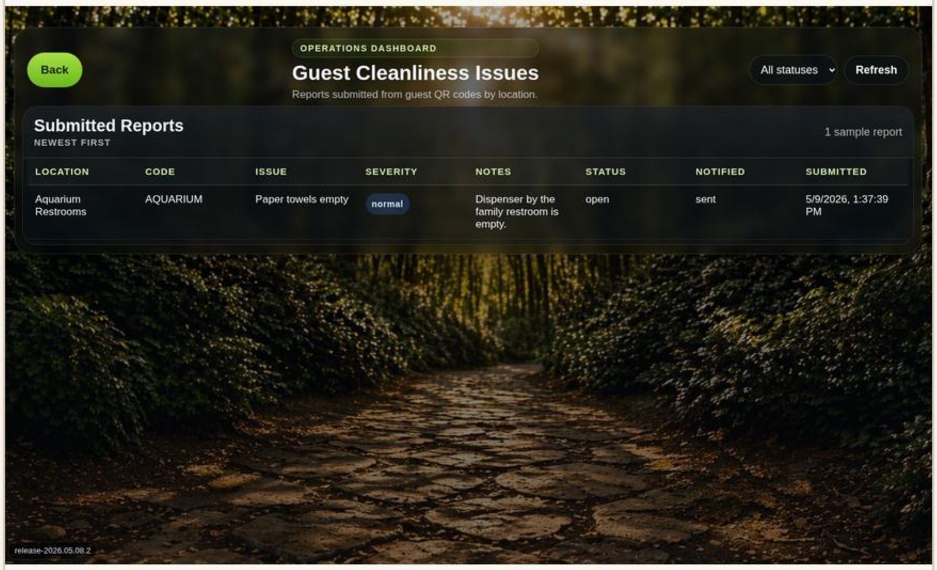
WHAT THIS SCREEN SHOWS

- Shows the clean, readable event display layout.
- Refresh behavior keeps event information aligned with backend data.
- When events exist, they can be tied back to areas so the right custodial coverage receives the right prompt.
- The empty state is useful because it makes clear when no event is currently scheduled.

Guest Issue Reporting

A guest report should become an operations signal, not a loose comment that someone has to manually route.

Guest-facing QR landing page with sample issue



Operations-side guest issue view

NOTIFICATION PATH

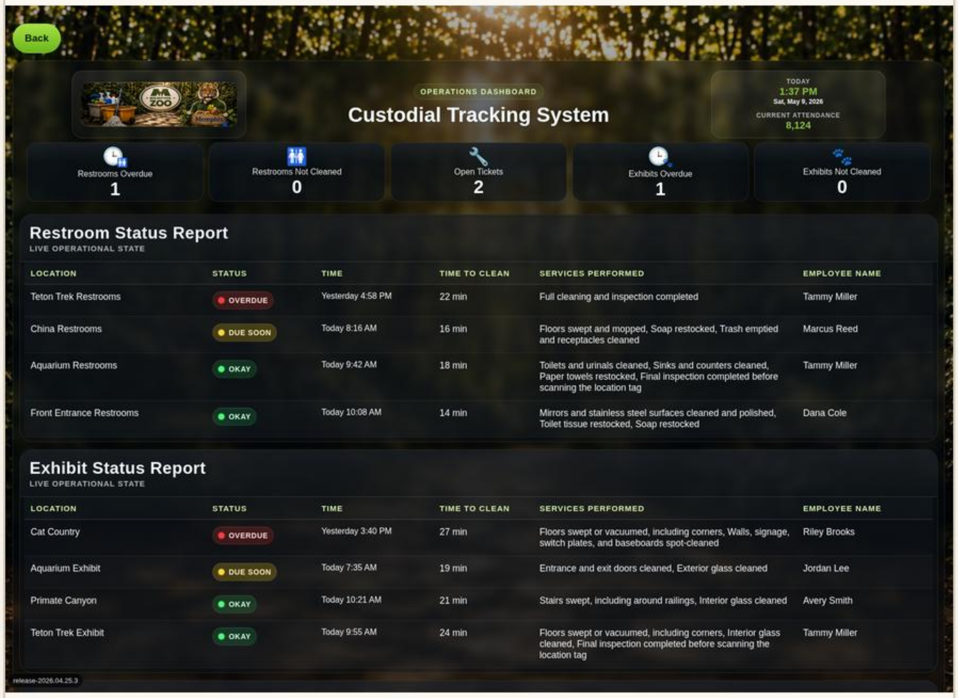
- Guest scans the location QR code and sees a simple landing/report page for that area.
- The sample issue appears on the guest page and on the operations-side issue screen so the path is easy to understand.
- Memphis Messenger notifies the assigned custodian and operations contact, following the same pattern as event, due soon, and overdue notifications.
- Managers can review submitted issues from the operations screen and close the loop after follow-up.

Why this matters

Guest-facing QR reporting turns a complaint into a routed task with a location, timestamp, and follow-up path.

Scan Workflow and Dashboard Visibility

The scan workflow gives custodians clear instructions and gives managers a PC view of what is okay, due soon, or overdue.



Custodian scan instructions

Manager / ops dashboard with realistic examples

WHAT THE WORKFLOW PROVES

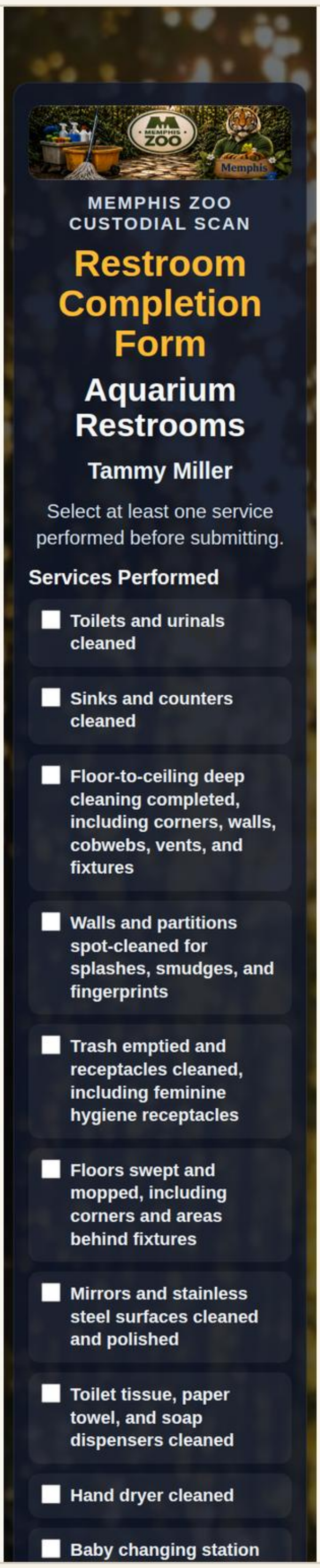
- Custodians scan the location tag, select their name, start cleaning, and scan again when finished.
- The dashboard converts scan activity into simple visibility: OKAY, DUE SOON, OVERDUE, and open tickets.
- Restroom and exhibit examples appear together, so managers can see the full custodial picture from a PC.
- This does not replace inspection; it tells managers when and where cleaned areas are ready to inspect.

Operational value

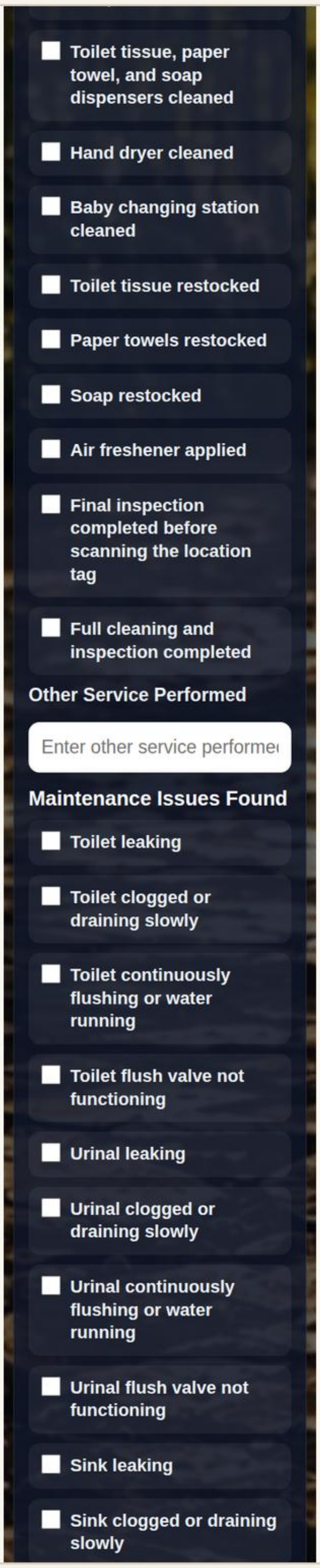
The phone screen guides the field workflow. The PC screen turns that field activity into inspection readiness and follow-up visibility.

Restroom Completion Form

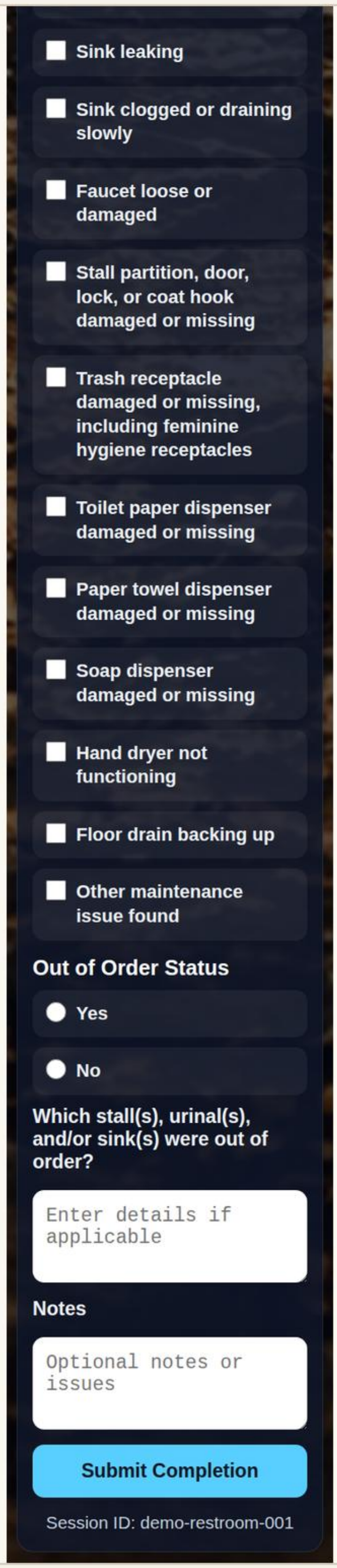
The long phone form is shown in three readable slices so the checklist, maintenance issue section, notes, and submit button are visible without wasting space on wallpaper.



Form start



Services and issues



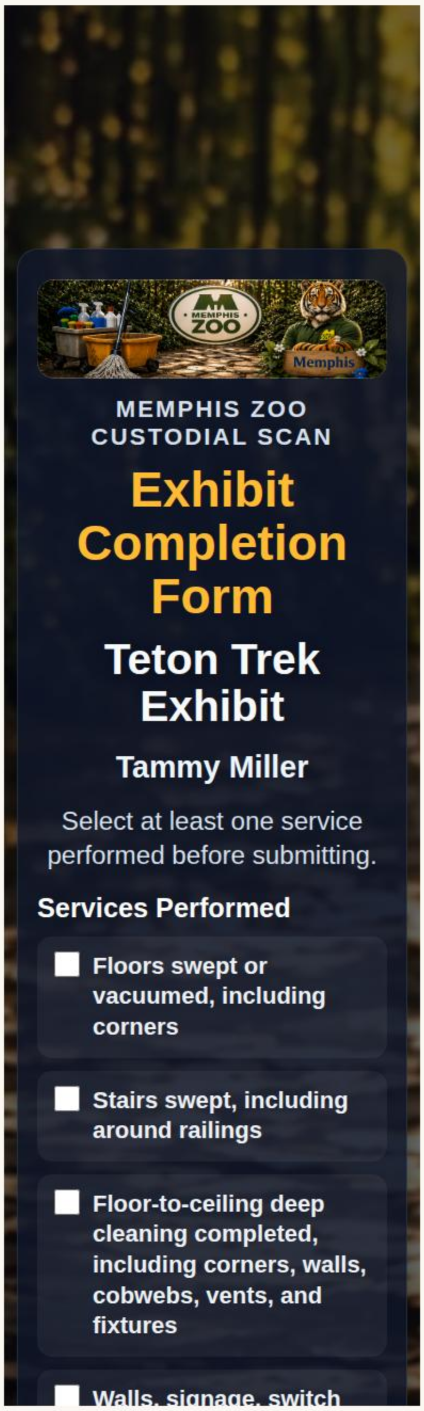
Notes and submit

WHY THE FORM MATTERS

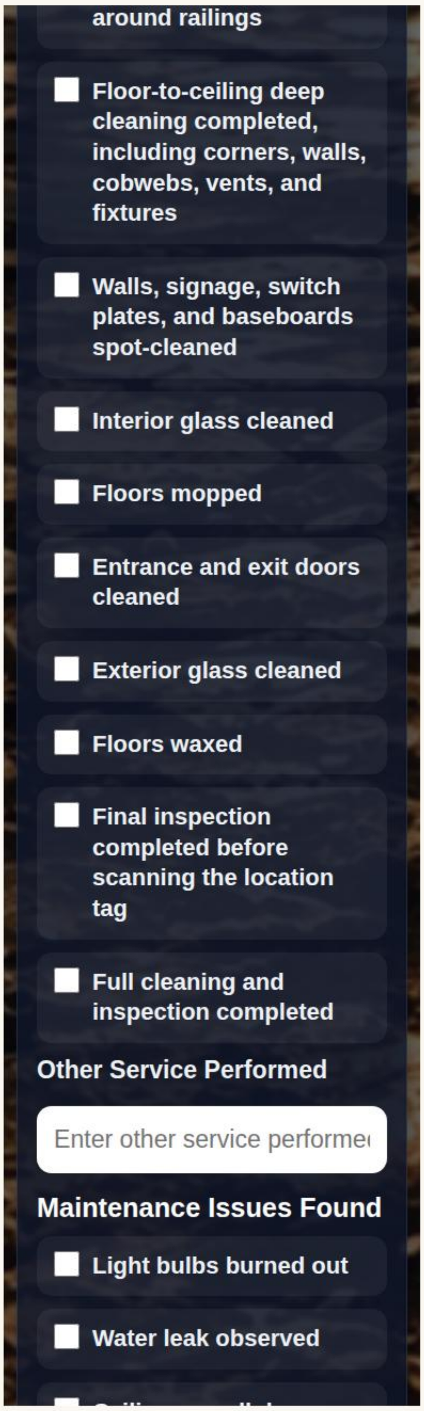
- Restroom checklist includes toilets/urinals, sinks/counters, mirrors, floors, trash, paper products, soap, odors, and final inspection.
- The same form captures fixture-level maintenance issues, out-of-order needs, and notes before submission.
- The final submission creates the record managers use for follow-up and inspection readiness.

Exhibit Completion Form

The long phone form is shown in three readable slices so the checklist, maintenance issue section, notes, and submit button are visible without wasting space on wallpaper.



Form start



Services and issues

Notes and submit

WHY THE FORM MATTERS

- Exhibit checklist includes floors, glass, stairs, railings, doors, signage, baseboards, cobwebs, trash, and final inspection.
- Maintenance issues and notes are captured before submission, just like the restroom workflow.
- The final submission creates the record managers use for follow-up and inspection readiness.

Actual Project Cost and Monthly Cost

The implementation budget is based on the attached equipment table. The phones are included as a real one-time equipment cost; there is still no phone carrier plan and no current monthly software cost.

ACTUAL EQUIPMENT PURCHASES

Item	Qty	Unit Cost	Subtotal
Moto G Phones	10	$39.99	**$399.90**
USB Charging Cords	2	$7.59	**$15.18**
Charging Station	1	$31.98	**$31.98**
Phone Holsters	10	$12.99	**$129.90**
NFC Tags	4	$13.99	**$55.96**
NFC Reader	1	$37.99	**$37.99**

Total Project Cost **$670.91**

Monthly cost clarification

The equipment above is the real implementation purchase total. The phones do not require a carrier subscription because they run on zoo Wi-Fi. Current monthly software cost remains $0/month for the frontend/backend services

CURRENT SERVICE COST

GitHub Pages	**$0/month current**	Hosts the static frontend pages.
Supabase	**$0/month current**	Current backend data layer remains on the free tier unless scale, backups, or policy needs change.
Render	**$0/month current**	Current API/backend hosting remains on the free tier unless scale or policy needs change.

EXPANSION MODEL

- Adding more locations mainly means adding more NFC/QR points and configuration records, not buying a new software package.
- Adding more phones means additional low-cost hardware and accessories, not a carrier bill when the phones stay on zoo Wi-Fi.
- The realistic ongoing cost is replacing broken phones, damaged/missing NFC tags, charging accessories, or optional cloud upgrades.
- The system avoids paying for unused CMMS modules such as parts inventory, purchasing, vendor management, or asset depreciation unless those are needed later.

Commercial Software Comparison

Most available systems cover pieces of this workflow. Matching the full combination usually requires multiple products or higher-tier plans.

MaintainX

Monthly: $20-$75 per user on published paid tiers.

Startup/equipment: implementation may be separate; phones/tablets, labels, and any kiosk setup are usually customer-side. Strong CMMS, but not this full zoo-specific workflow out of the box.

OrangeQC

Monthly: about $250-$500, plus add-ons and extra inspectors.

Startup/equipment: QR shortcuts exist, but field devices, printed codes/tags, and rollout work are still separate. Good inspection/ticket fit, not the whole custom operations hub.

UpKeep

Monthly: about $24-$55 per user on published tiers.

Startup/equipment: published implementation/training packages range from about $500 to $5,000; field devices/labels are still separate. Broad platform, more than custodial needs.

Why the monthly cost jumps

Commercial tools often become expensive once several paid users, QR workflows, offline mode, analytics, implementation help, request portals, API access, or multiple modules are needed.

Initial setup difference

Many systems do not include phones, tablets, printed QR labels, NFC tags, kiosk setup, data cleanup, or local rollout work in the monthly subscription. Those are separate startup costs.

Commercial Software Comparison

Most available systems cover pieces of this workflow. Matching the full combination usually requires multiple products or higher-tier plans.

Limble CMMS

Monthly: quote/calculator model on public page.

Startup/equipment: QR/work request features exist, but pricing and rollout costs depend on quote; devices and labels are typically separate. Strong CMMS, broader than this use case.

SafetyCulture

Monthly: team inspection/checklist platform pricing.

Startup/equipment: mobile devices, printed QR codes, and rollout configuration are outside the base software concept. Good inspections, but not schedule + messenger + events + Memphis AI.

Is there one system this comprehensive?

Public vendor offerings cover important pieces, but one ready-made system combining zoo custodial scans, guest QR issues, events, schedule reassignment, Messenger, reminders, and Memphis AI is not obvious.

Cost conclusion

The custom path wins on cost because current monthly software cost is $0, startup hardware is minimal, and future spending is limited to replacement phones, replacement tags, or optional cloud upgrades.

Real-World Operating Pattern

Comparable venues already use QR reporting, smart restrooms, mobile inspections, guest feedback, and facility dashboards. The Memphis Zoo value is combining those proven patterns into one low-cost custodial workflow.

COMPARABLE VENUE USES

- Airports: ELERTS describes airport restroom QR reporting where guests scan a code, submit details or a photo, and the location-tagged report goes directly to airport maintenance.
- Airports: TRAX describes Kansas City International Airport restrooms using guest feedback, connected dispensers, cleaning alerts, mobile janitorial workflows, and performance data.
- Large venues: TRAX markets restroom intelligence for airports, arenas, and stadiums, which are high-traffic facilities with the same need for fast cleaning visibility.
- Hotels: Bitly and The QR Code Generator both describe QR housekeeping requests that reduce phone calls, route guest needs, and give staff a clearer digital queue.
- Visitor attractions: theme parks and aquariums face the same operating problem as a zoo: dispersed restrooms, high guest volume, event spikes, and teams that need location-specific prompts.

WHAT THIS SUPPORTS

- The individual ideas are not experimental: QR intake, mobile checklists, guest feedback, smart restroom data, dashboards, and alerts already appear in real facility operations.
- The Memphis Zoo system adapts those ideas to custodial work instead of buying an oversized enterprise platform.
- The custom advantage is the combined workflow: schedule ownership, NFC/QR proof, guest reports, events by area, reminders, Messenger notifications, manager visibility, and Memphis AI in one place.

SAFE COMPARISON

The best claim is simple: comparable facilities validate the direction, while this implementation is tailored to the zoo’s exact day-to-day operation and current $0/month software cost.

SOURCES CHECKED

ELERTS airport restroom QR reporting; TRAX Kansas City International Airport case study and TRAX facility categories; Janitorial Manager public scan QR codes; Bitly hotel housekeeping QR requests; The QR Code Generator hotel housekeeping QR requests; vendor examples for smart washroom use in high-traffic venues.